

Simple Linear Regression

Jinsong Chen

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i Chapter overview

This chapter reads a fitted line. Starting from a reported claim about education and income, we state an occupation-level question, plot the records, and fit a straight line to them. We read the slope, intercept, fitted values, and residuals; see how least squares chooses the line and how the slope is covariance divided by predictor variation; and describe how much variation the line leaves. We then declare a separate process question, derive the slope's standard error and interval, and watch fitted slopes vary across repetitions of a known process. We distinguish a mean response from one new outcome, check form, spread, influence, and support, and close with Galton's heights of fathers and sons, where the line in standard units explains regression toward the mean. By the end of the chapter you will be able to fit and read a simple regression in R, state what its descriptive and model-based answers concern, and recognize regression toward the mean in a before-and-after claim. An appendix develops the mathematics further.

Suppose a report says, "Another year of education increases income by about 900 dollars." Before examining its significance, ask whose education and whose income were measured. The same number could refer to individuals, households, or occupational averages. Those are different comparisons.

Our question is deliberately narrower:

Among the 102 recorded Canadian occupations, how did occupation-average income differ across the supported range of occupation-average education in 1971, as summarized by a fitted straight line?

One row represents an **occupation**, not a worker. An occupation with higher average education need not contain the same people as an occupation with lower average education. A comparison between their averages does not describe what would happen to one person's income after another year of schooling. Aggregate and individual relationships can differ substantially (Robinson, 1950).

Chapter 5 connected covariance and correlation to recorded values and made mean and correlation inference executable in R. Chapter 6 fitted group means through indicator coefficients in a model formula. Here we develop the fitted line for a continuous predictor: how its slope is calculated, what its uncertainty requires, and which comparisons and predictions the records support.

1 A Question About Occupations and a Line Through Their Records

1.1 Begin with the unit and the question

The data are distributed as `carData::Prestige` (Fox et al., 2022; Fox & Weisberg, 2019). The package documentation attributes education and income to the *Census of Canada* (1971), volume 3, part 6, pages 19-1 to 19-21. We read the package documentation as a source record, not as evidence that these occupations form a probability sample. The chapter’s calculations are our reanalysis, not statements attributed to an original research paper.

Table 1

Record, measurements, and scope for the focal analysis.

Feature	Meaning in this chapter
Unit and record count	102 occupations; each occupation receives equal weight
Education, x	Average years of education of occupational incumbents in 1971
Income, y	Average income of incumbents, divided by 1,000; thousand 1971 Canadian dollars
Missingness for the fit	No missing education or income values; no modeled rows excluded
Observed support	Education: 6.38–15.97 years; income: 0.611–25.879 thousand dollars
Descriptive estimand	Least-squares slope summarizing these 102 fixed occupation records
Limits of the source record	No probability-sampling basis or individual-level analysis established here

Equal weighting matters. A small occupation and a large occupation contribute one row each. Without suitable counts and a different estimand, this analysis does not describe an average worker’s experience. Historical dollars also cannot be read as contemporary purchasing power.

1.2 Look before fitting

```
prestige <- carData::Prestige
prestige$income_k <- prestige$income / 1000
library(ggplot2)
```

The package-qualified name `carData::Prestige` selects the supplied dataset. Dividing income by 1,000 changes its units while preserving every occupation. The Computer Lab supplies installation and operating practice if needed; the occupation example requires no separate data file.

```
ggplot

```

prestige, aes(education, income_k)) +
 geom_point(shape = 1, size = 2) +
 labs(x = "Average education (years)",
 y = "Average income (thousand 1971 dollars)"))

```


```

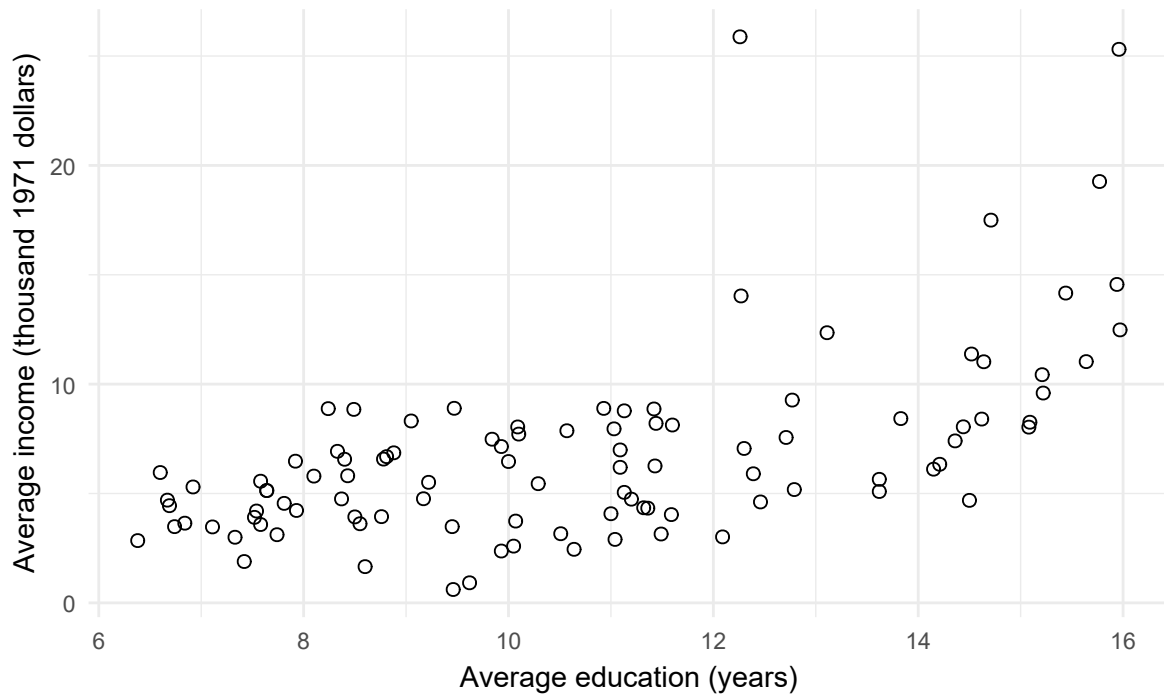


Figure 1: Education and income in the 102 occupation records, before fitting a line.

The scatterplot gives the line a job: summarize an overall upward pattern amid substantial variation. It also shows why the line cannot describe every occupation equally well. General managers, for example, have an average education of 12.26 years and an average income of 25.879 thousand dollars; university teachers have corresponding values of 15.97 and 12.480. More education does not order every pair of occupations by income.

The scatterplot motivates fitting a straight-line summary. Save that fit as `model`, and keep its coefficients in `b` for calculations that follow.

```
model <- lm(income_k ~ education, data = prestige)
b <- coef(model)
coef(model)
#> (Intercept)      education
#> -2.8535855824  0.8988128418
```

The formula puts the outcome to the left of the tilde and the predictor to its right. The command finds coefficients. It does not choose a population, establish a causal design, or justify a confidence interval.

2 Read the Fitted Line and Understand How OLS Finds It

2.1 Fitted values, slopes, and residuals

A straight-line summary assigns each observed predictor value x_i a **fitted value** $\hat{y}_i$:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \quad e_i = y_i - \hat{y}_i.$$

The subscript i indexes a record. The hat marks a fitted quantity. The **residual** e_i is observed income minus fitted income. A positive residual places an occupation above the line; a negative residual places it below. A residual is calculated after fitting and is not the same as an unobservable error from a population model.

For the occupation records, the line is

$$\widehat{\text{income}} = -2.8536 + 0.8988 \text{ education}.$$

Income is still in thousands of dollars. The slope says that the fitted income differs by 0.8988 thousand dollars, about 899 dollars, for a one-year difference in occupation-average education. A two-year difference along this line corresponds to $2\hat{\beta}_1 \approx 1.7976$ thousand dollars.

For university teachers, use the fitted coefficients before rounding:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(15.97) \approx 11.5005, \quad e = 12.4800 - \hat{y} \approx 0.9795.$$

Their recorded average income is about 980 dollars above the fitted value. The printed regression equation abbreviates the coefficients; the stored model retains the digits needed for these calculations. R reproduces the fitted value and signed residual directly:

```
teacher <- prestige["university.teachers", ]
teacher_fit <- predict(model, teacher)
c(fitted = unname(teacher_fit),
  residual = teacher$income_k - unname(teacher_fit),
  at_12 = unname(predict(model, data.frame(education = 12))))
#>      fitted      residual      at_12
#> 11.5004555017  0.9795444983  7.9321685196
```

For an occupation averaging 12 years of education, the fitted value is 7.9322 thousand dollars. `predict()` applies the saved line to the named education value; it does not refit the model.

Table 2 sets the slope’s reading beside two sentences a report might also write about it.

Table 2

Three readings of one slope; only the first is a comparison these records make.

A sentence about the slope 0.8988	What it compares	Do these records support it?
Occupations whose average education is one year higher have fitted average income about 899 dollars higher	Different occupations, along the fitted line	Yes, as a description of the 102 records over 6.38–15.97 years
If an occupation’s average education rose by one year, its average income would rise by about 899 dollars	One occupation before and after a change	No: it needs records of occupations over time and a design that isolates the change
Another year of schooling raises a worker’s income by about 899 dollars	One person before and after more schooling	No: it needs records of individuals and a causal design; occupation averages cannot supply either

Only the first sentence compares what the records compare: occupations that differ in average education. The second describes a change within one occupation over time and the third a change within one person. The records observe neither, and a comparison between occupation averages cannot stand in for either one (Robinson, 1950). The slope supports the first reading and is silent about the other two.

The intercept places the line vertically: it is its fitted income at education zero. Zero is outside the observed range, so the negative intercept is not evidence that any recorded occupation has negative average income. Centering education at 12 would change the intercept to 7.9322 while leaving the slope and all fitted values unchanged.

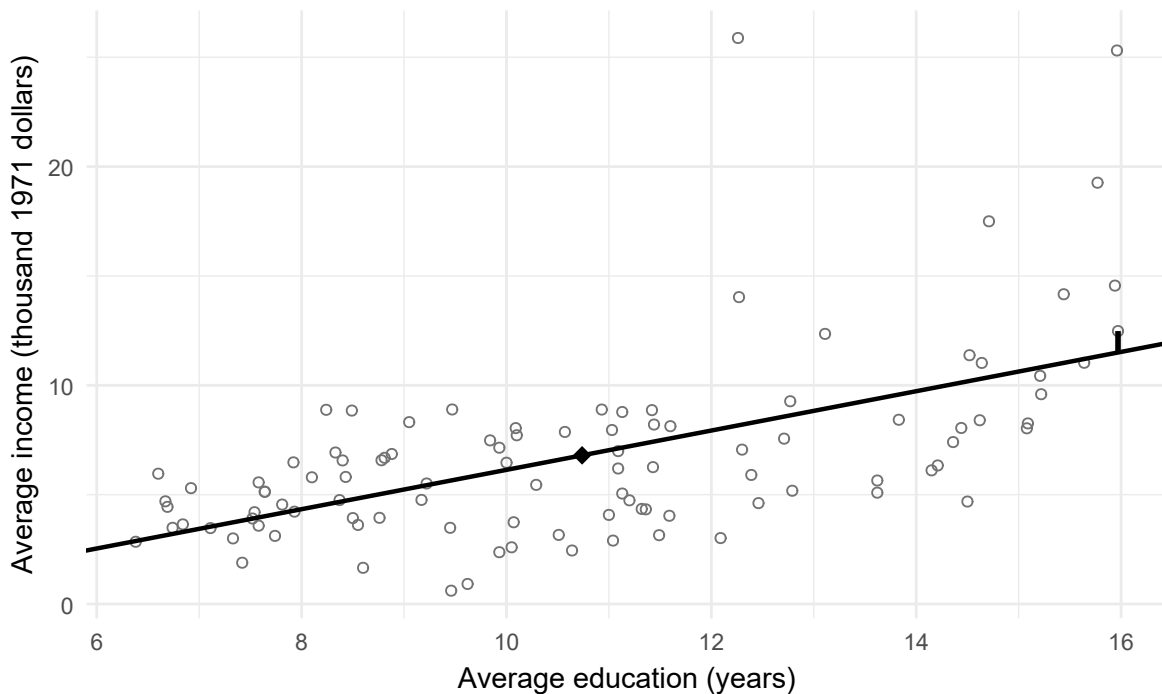


Figure 2: The least-squares line passes through the mean point; the marked vertical segment is the university-teacher residual.

2.2 Why this particular line?

Ordinary least squares (**OLS**) chooses the intercept and slope that minimize the **residual sum of squares**:

$$RSS = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

The summation sign says to add the squared residual from every record. Squaring prevents positive and negative deviations from cancelling and gives large deviations more influence on the criterion. “Best” here means best by this criterion among straight lines with an intercept, not best for every research purpose.

Two centered sums lead directly to the solution:

$$S_{xx} = \sum_i (x_i - \bar{x})^2, \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}),$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{s_{xy}}{s_x^2} = r \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Here $\bar{x}, \bar{y}$ are means; s_x, s_y are sample standard deviations; s_{xy} is sample covariance; and r is Pearson's correlation. The variance and covariance both divide their centered sums by $n - 1$, so that divisor cancels in their ratio. The slope has income-per-education units; correlation has no units.

For these data,

$$\hat{\beta}_1 = \frac{675.8041}{751.8852} \approx 0.8988, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx -2.8536.$$

Chapter 5's covariance and correlation matrices give the same calculation in a reusable form. For the same 102 complete education-income pairs, in education-income order,

$$\mathbf{S} = \begin{pmatrix} 7.4444 & 6.6911 \\ 6.6911 & 18.0279 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} 1 & .5776 \\ .5776 & 1 \end{pmatrix}.$$

The covariance matrix records education variance in years squared, income variance in squared thousand dollars, and their covariance in thousand-dollar-years. Dividing covariance by education variance cancels the common $n - 1$ divisor and gives the slope's thousand-dollars-per-year units. Correlation removes units; multiplying it by the income SD divided by the education SD restores those same slope units. Computed from the unrounded entries, both routes return 0.8988: they reproduce the centered-sum calculation rather than define a different fitted line. The Computer Lab carries out the centered sums and both matrix routes from the records.

The shared records matter as much as the formula. Entries formed by separate pairwise deletion can describe different sets of occupations; they need not reconstruct a regression fitted on one common set. No such missingness occurs in these two columns.

Positive covariance produces a positive slope, provided education varies at all. If every occupation had the same education, $S_{xx} = 0$ and a slope could not be identified.

The fitted line passes through $(\bar{x}, \bar{y})$, because substituting $\bar{x}$ into the intercept formula gives $\bar{y}$. With an intercept, the residuals also sum to zero and their products with x sum to zero. Otherwise the line could be shifted or tilted to reduce RSS. These are algebraic properties of OLS, not tests of independence or linearity. Full normal-equation and projection-matrix derivations are retained in **Appendix A**.

2.3 A tiny calculation you can inspect completely

Three teaching records, $(x, y) = (1, 2), (2, 4), (3, 3)$, show the whole least-squares calculation at once. Their means are $\bar{x} = 2$ and $\bar{y} = 3$. Table 3 lays out the centered values, their squares and products, the fitted values and the residuals; its last row sums each column.

Table 3

The complete least-squares calculation for three records; the column sums give S_{xx} , S_{xy} and RSS .

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$	$\hat{y}$	e	e^2
1	2	-1	-1	1	1	2.5	-0.5	0.25
2	4	0	1	0	0	3.0	1.0	1.00
3	3	1	0	1	0	3.5	-0.5	0.25
Sum		0	0	2	1	9.0	0.0	1.50

The sums give $S_{xx} = 2$ and $S_{xy} = 1$, so

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{1}{2} = 0.5, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3 - 0.5(2) = 2.$$

The fitted values 2.5, 3.0 and 3.5 lie on this line. The residuals sum to zero, as the OLS properties above require, and their squares sum to $RSS = 1.5$. No normality assumption is needed for these calculations. A stochastic model becomes necessary when we ask about repeated-sample uncertainty.

2.4 Compare candidate lines with the OLS minimum

The least-squares criterion also lets us compare the fitted line with alternatives. To keep the comparison transparent, make every candidate line pass through $(\bar{x}, \bar{y})$ and vary only its slope. For such a line,

$$RSS(b_1) = RSS(\hat{\beta}_1) + S_{xx}(b_1 - \hat{\beta}_1)^2.$$

The added term is zero only at the OLS slope and positive for every other slope when $S_{xx} > 0$. The following table checks five deliberately chosen lines on the occupation records. These are alternative summaries of the same records, not repeated samples.

Table 4

Candidate lines through the mean point and their residual sums of squares.

Intercept	Slope	RSS
3.5892	0.2988	1484.0707
0.3678	0.5988	1281.0617
−2.8536	0.8988	1213.3920
−6.0750	1.1988	1281.0617
−9.2964	1.4988	1484.0707

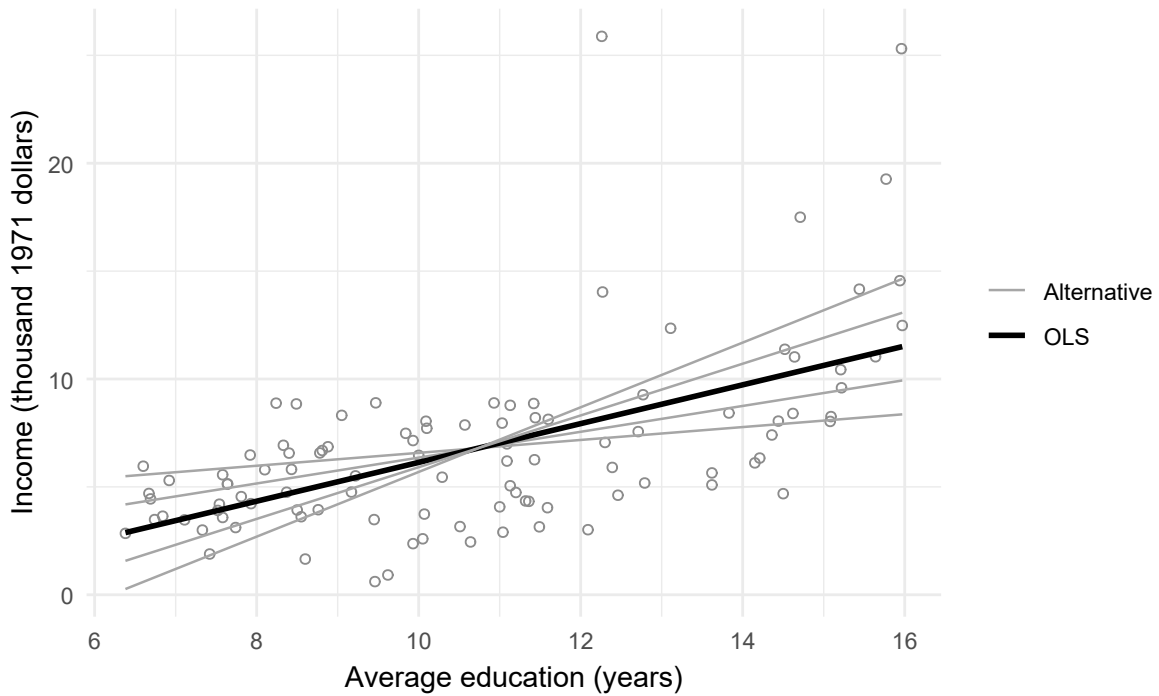


Figure 3: Five candidate lines through the sample mean. The OLS line has the smallest residual sum of squares.

Every alternative has a larger RSS than the OLS line, and slopes the same distance above and below 0.8988 add the same excess, as the identity predicts. The comparison also explains why high-income occupations can exert substantial influence: squaring gives large vertical discrepancies extra weight. That is a feature of the criterion, not evidence that those occupations are invalid.

2.5 Centering changes the reference point, not the line

The fitted equation can be written

$$\hat{y}_i = \bar{y} + \hat{\beta}_1(x_i - \bar{x}).$$

Centering education at its observed mean makes the intercept equal to $\bar{y} = 6.7979$ thousand dollars, the fitted income at average education. It leaves the slope, fitted values, residuals, RSS, and R^2 unchanged. Centering at 12 instead makes the intercept the already computed fitted value 7.9322. Thus centering can make the intercept interpretable without changing the comparison represented by the line. The Computer Lab refits the centered line and checks that every fitted value is unchanged.

Changing income from thousands of dollars to dollars would multiply the intercept, slope, residual SD, and coefficient SEs by 1,000, while leaving t statistics, p values, correlation, and R^2 unchanged. A unit change is not new evidence or a stronger relationship.

2.6 How much variation remains?

A fitted line can show a clear linear trend and still leave large differences between occupations unexplained by its fitted values.

For OLS with an intercept,

$$TSS = \sum_i (y_i - \bar{y})^2 = RegSS + RSS, \quad RegSS = \sum_i (\hat{y}_i - \bar{y})^2.$$

For the occupation records, $1820.8134 = 607.4214 + 1213.3920$. This is a decomposition of squared deviations, in squared thousand-dollar units. It holds because the cross-product of fitted deviations and residuals is zero.

The coefficient of determination is

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{1213.3920}{1820.8134} \approx .3336.$$

The fitted line accounts for 33.3599% of the variation in recorded income around its mean, according to this squared-deviation measure. “Accounts for” or “explains” in this algebraic sense does not mean causes. For simple regression with an intercept, $R^2 = r^2$: $.5776^2 \approx .3336$. The identity need not hold for other models or a no-intercept fit.

The residual standard deviation is

$$s_e = \sqrt{\frac{RSS}{n-2}} = \sqrt{\frac{1213.392}{100}} = 3.4834.$$

Its units are thousand dollars, the same as income. Fitting an intercept and slope imposes two independent residual constraints, leaving $n - 2$ residual degrees of freedom. Under the constant-variance model introduced next, s_e estimates the error standard deviation. Without that model it remains a scaled measure of residual scatter; it is not the standard error of the slope or a guaranteed typical prediction error in new records.

`sigma()` returns the residual SD of a fitted model, and `summary()` reports its R^2 among the other fit summaries:

```
c(residual_SD = sigma(model), R_squared = summary(model)$r.squared)
#>  residual_SD      R_squared
#> 3.4833777071 0.3335989192
```

Here and below, SE denotes an estimated standard error. The final row anticipates the separately declared process in the next section; it is not uncertainty about the already computed finite-record slope.

Table 5

Different summaries have different units and jobs.

Quantity	Occupation result	What it describes
Slope	0.8988	Change in fitted thousand-dollar income per education year
Residual SD	3.4834	Remaining income scatter on the outcome scale
R^2	.3336	Proportion of centered squared variation accounted for in these records
Classical slope SE	0.1270	Model-based repetition variability of the slope, under the next assumptions

3 Make and Read a Model-Based Inference

3.1 Announce the inferential extension before its interval

The finite-record slope is not waiting to be discovered in a larger population: we have computed it for all 102 records named in the focal question. A confidence interval answers an additional question.

For a **separable teaching extension**, suppose occupations' outcomes arose independently from a process with conditional mean

$$E(Y_i | X_i = x_i) = \beta_0 + \beta_1 x_i.$$

The estimand β_1 is now a constant conditional-mean slope in that hypothetical process, not the fixed slope of these 102 records. Repetition means generating new outcomes from that process at the stated predictor values and refitting. It does not mean that the documented occupations were randomly sampled. Changing the estimand changes the question–answer pair.

Write the process as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad E(\varepsilon_i | \mathbf{X}) = 0, \quad \text{Var}(\varepsilon_i | \mathbf{X}) = \sigma^2.$$

For the exact classical t procedure, errors are independent and conditionally normal with this common variance, and $S_{xx} > 0$. The x_i values need not be normal; neither must the marginal distribution of Y . It is the conditional error distribution, not the histogram of every variable, that appears in the model. Independence is supported by how records arise, not by a good-looking residual plot.

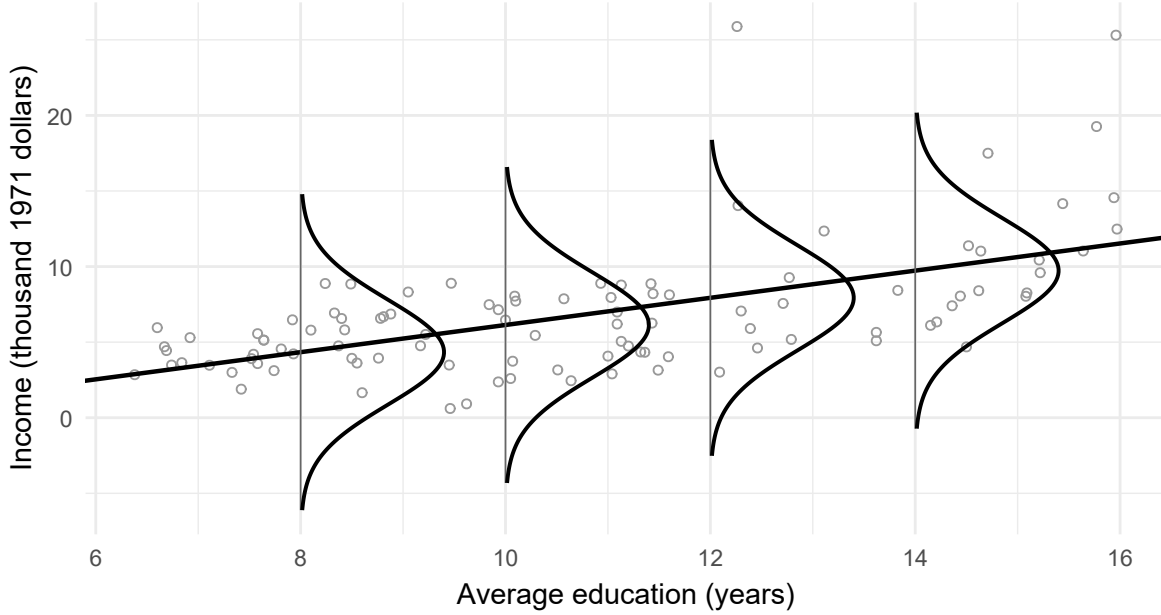


Figure 4: What the process conditions claim, drawn on the occupation records: at each education value, a Normal distribution of incomes centred on the line, with the same spread everywhere.

Figure 4 draws what these conditions claim, using the fitted line and the residual SD, 3.4834, in place of the unknown β_0 , β_1 and σ . At every education value the model places a Normal distribution of incomes centred on the line, with one common spread. The records do not look like that everywhere: at higher education they spread more widely, and several incomes lie far beyond the upper tails, which Section 6 examines. At 8 years the curve reaches below zero, a first sign that a Normal model for a nonnegative outcome is an approximation, which Section 5 meets again.

To see where slope uncertainty comes from, substitute the process equation into the OLS slope formula. Here $\mathbf{X}$ denotes the predictor values on which we condition. Since $\sum_i (x_i - \bar{x}) = 0$,

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x}) \varepsilon_i}{S_{xx}}.$$

The fitted slope differs from the process slope because the realized errors need not cancel when weighted by distance from mean education. Independence makes their covariance terms zero; common variance gives

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2 \sum_i (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

This variance identity does not require normal errors. Normality enters the exact t reference. Taking the square root and estimating σ by s_e gives the slope standard error:

$$SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{S_{xx}}} = \frac{3.4834}{\sqrt{751.8852}} = 0.127,$$

$$\hat{\beta}_1 \pm t_{.975, n-2} SE(\hat{\beta}_1) \approx 0.8988 \pm 1.984(0.127) \approx [0.6468, 1.1508].$$

The critical value 1.984 is the 97.5th percentile of a t distribution with 100 degrees of freedom. More error variation raises the SE; greater predictor spread lowers it when the other quantities are held fixed. More observations usually improve precision through their contribution to S_{xx} , but dependent observations do not supply independent information merely by increasing the row count.

The null test $H_0 : \beta_1 = 0$ uses $t = \hat{\beta}_1 / SE(\hat{\beta}_1) \approx 7.0753$ and a two-sided $p < .0001$. This is evidence against that zero-slope model under the stated procedure. It is not the probability that the null is true, the probability that the data occurred by chance, evidence of causation, or a measure of practical importance. The interval describes the range produced by this uncertainty method, not a 95% probability assigned to the fixed parameter after observing this interval. The next section makes the repeated-sample interpretation visible before prediction is introduced.

The coefficient table gives the estimate, its SE, and the test; the interval call returns the corresponding two-sided limits. The function's default confidence level is .95.

```
coef(summary(model))
#>               Estimate Std. Error    t value    Pr(>|t|)
#> (Intercept) -2.8535855824 1.4070392470 -2.028078171 4.521163138e-02
#> education    0.8988128418 0.1270354112  7.075293679 2.079192410e-10
confint(model, "education")
#>           2.5 %      97.5 %
#> education 0.6467782042 1.150847479
```

The assumptions are a proposed justification, not facts created by the formula. Curvature, changing spread, or dependence can invalidate the exact coverage claim. Section 6 examines these conditions on the occupation records.

3.2 Connect the Pearson and slope tests

Chapter 5 tested a zero Pearson correlation. The simple regression now lets us connect that familiar statistic to the slope calculation. For an OLS line with an intercept, fitted to the same complete pairs,

$$t_{\text{slope}} = \frac{\hat{\beta}_1}{s_e / \sqrt{S_{xx}}} = r \sqrt{\frac{n-2}{1-r^2}} = t_{\text{Pearson}},$$
$$F_{\text{regression}} = t_{\text{slope}}^2 = \frac{R^2}{1-R^2}(n-2).$$

The first identity follows by substituting the covariance expression for the slope and using $RSS = (1-r^2)TSS$. The second compares the intercept-only mean model with the intercept-and-slope model: their numerator degrees of freedom differ by one. The signed t retains the slope's direction; squaring it gives the nondirectional F statistic.

For the occupation line, Chapter 5's Pearson test and the slope test both give $t = 7.0753$ on 100 df, and the F statistic that `summary()` prints is $F(1, 100) = 50.0598 = t^2$; the matched two-sided t and upper-tail F p-values are below .0001. These entries express the same one-parameter comparison, not three independent pieces of evidence. The Computer Lab reproduces all three from the fitted model.

Equality of the statistics is algebraic. Their exact classical references use the independent, constant-variance Normal error model already stated. For randomly sampled bivariate-Normal pairs, a zero population correlation is also equivalent to a zero linear conditional-mean slope. The documented occupation records provide no probability-sampling warrant: here the Pearson call reproduces the declared process calculation numerically, rather than adding a new population-correlation claim.

Match the intercept, rows, weights, and covariance convention before comparing outputs. An HC3 slope test uses a different covariance estimate; it need not equal the ordinary ANOVA F calculation. Section 6 uses that change to examine uncertainty while keeping the fitted line fixed.

3.3 A binary predictor: the two-group comparison as a line

A predictor need not be continuous. Chapter 4's invented feedback study recorded a practice schedule for each of its 24 pupils, twelve under massed practice and twelve under spaced practice, with a score out of 20; Chapter 6 fitted the same scores by feedback condition with a model formula. Code the schedule as $x_i = 0$ for massed and $x_i = 1$ for spaced, and fit the same kind of line. With only two predictor values, the least-squares line passes through the two group means (Figure 5): the intercept is the massed mean and the slope is the difference of means,

$$\hat{\beta}_0 = \bar{y}_0 = 8, \quad \hat{\beta}_1 = \bar{y}_1 - \bar{y}_0 = 12 - 8 = 4.$$

The residual SD is the pooled SD of the equal-variance procedure, $s_e = 4.156$ on $24 - 2 = 22$ df, and the slope SE formula becomes $SE(\hat{\beta}_1) = s_e \sqrt{1/n_0 + 1/n_1} = 4.156 \sqrt{1/12 + 1/12} = 1.6967$, so $t = 4/1.6967 = 2.3575$ on 22 df with $p = .0277$ and interval $[0.4813, 7.5187]$, which the Computer Lab reproduces with `t.test()`. These are exactly the equal-variance two-sample t test and interval for a difference in means (Chapters 3 and 5): a regression on a 0/1 predictor is that procedure written as a line. The matched statistics hold as well, with $R^2 = .2017$ and $F(1, 22) = 5.5579 = t^2$.

```
score <- c(4, 5, 7, 8, 4, 5, 7, 8, 6, 7, 9, 10, 10, 11, 13, 14,
          8, 9, 11, 12, 16, 17, 19, 20)
spaced <- rep(rep(c(0, 1), each = 4), 3)
two_groups <- lm(score ~ spaced)
coef(summary(two_groups))
#>               Estimate Std. Error    t value    Pr(>|t|)
#> (Intercept)         8 1.199747448  6.668070028 1.053846789e-06
#> spaced              4 1.696699113  2.357518767 2.770679048e-02
```

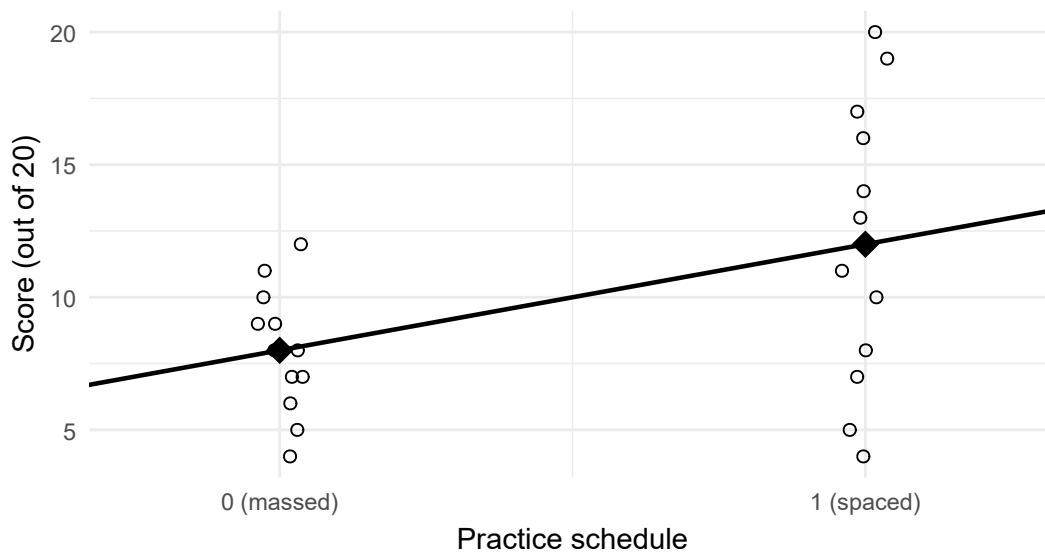


Figure 5: With a 0/1 predictor, the least-squares line joins the two group means: the intercept is the massed mean and the slope the difference in means.

Welch's unequal-variance test (Chapter 3) uses the separate group SDs, 2.3741 and 5.3767, and gives 15.1322 df and $p = .0323$. The one-line regression assumes a common error variance; that

is the model's assumption, not a fact about the pupils. The HC3 covariance of Section 6 keeps the fitted line and gives a slope SE of 1.7721 under the same independence assumption. A factor with more than two levels needs more than one 0/1 column; Chapter 9 develops that coding, its contrasts, and the equivalence with the analysis of variance of Chapter 4.

4 See Sampling Uncertainty Through Repetition

4.1 A known-truth line and one fitted sample

Generate 20 outcomes at predictor values equally spaced from 6 to 16:

$$Y_i = 2 + 0.5x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1).$$

These units and values are a teaching construction, not occupation records. The true intercept is 2, true slope is 0.5, and true error SD is 1.

```
set.seed(123)
known_data <- data.frame(x = seq(6, 16, length.out = 20))
known_data$y <- 2 + 0.5 * known_data$x + rnorm(20, sd = 1)
known_model <- lm(y ~ x, data = known_data)
```

`set.seed()` fixes the random start, so the same 20 errors are drawn each time, and `rnorm()` draws them with SD 1. Table 6 reports the fitted coefficients, their SEs and the residual SD of `known_model`.

Table 6

One fitted sample from a process with known coefficients and error SD.

Quantity	Estimate	Model SE
Intercept	2.4770	0.8362
Slope	0.4695	0.0733
Residual SD	0.9945	

The last cell is blank because residual SD is a scale estimate, not a coefficient with a coefficient standard error.

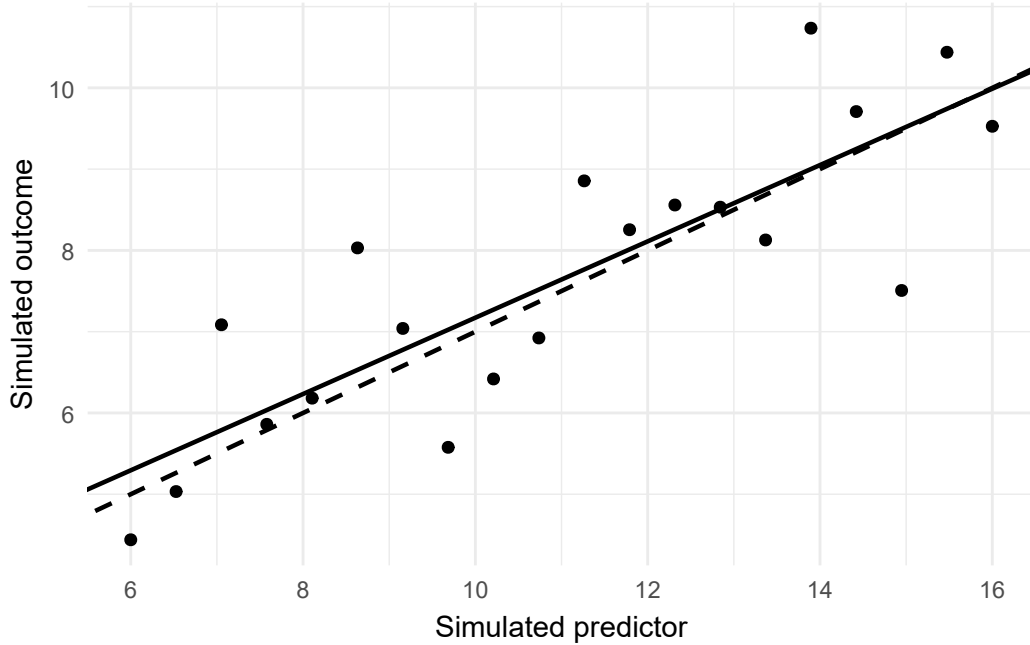


Figure 6: Known and fitted lines for one simulated dataset. The true process is programmed; the fitted line responds to the realized errors.

The fitted slope and intercept need not equal their programmed values in one sample. The unobserved error is the deviation from the true mean; the residual is the observed deviation from the fitted line. Even when the errors are independent, fitted residuals are linked by their zero-sum and zero-covariance constraints. A residual plot is an imperfect diagnostic window on error behavior, not a direct sample of observed true errors.

4.2 Repeated samples and the sources of slope precision

Recall the generate–estimate–repeat route from Chapters 5 and 6. The next experiment holds 40 predictor values fixed and regenerates only the errors, 2,000 times. The study size is 40; the repetition count is 2,000. Increasing the latter refines the simulation’s description of the same study-level procedure. It records slopes, estimated standard errors, and whether each classical interval covers the known slope 0.5.

For one repetition, generate new independent errors, add them to the same 40 process means, fit a line, and record its slope and estimated SE. Check whether its interval contains the programmed slope, then start the next repetition with new errors. Holding the predictor grid fixed isolates outcome variation; it does not resample the historical occupations. The Computer Lab provides the complete loop and a small change in the repetition count.

Table 7*Slope behavior across 2,000 repetitions of the known process.*

Quantity	Value
Mean estimated slope	0.5018
SD of estimated slopes	0.0528
Mean estimated SE	0.0531
Theoretical slope SD	0.0534
Fraction of intervals covering 0.5	.9455

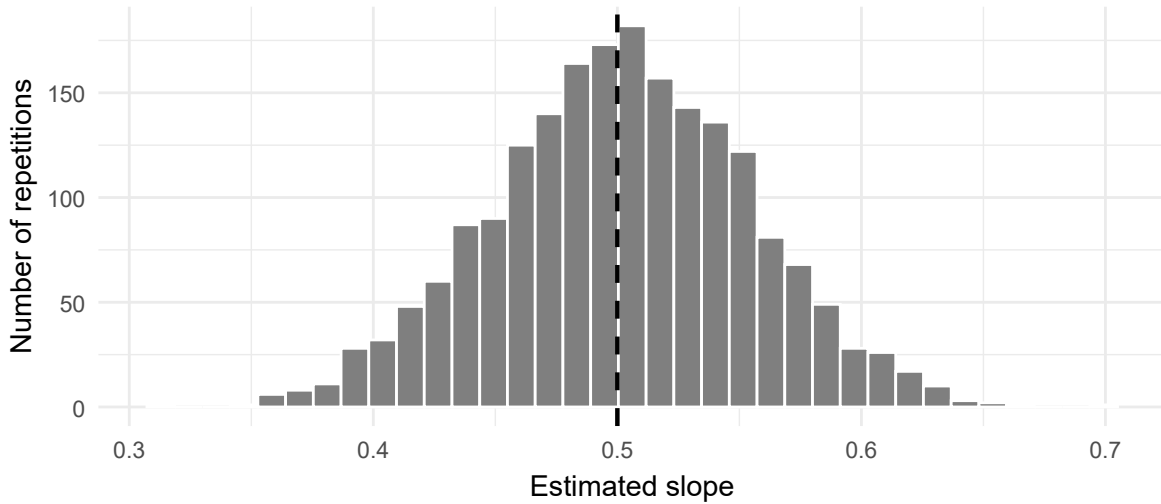


Figure 7: Slopes in 2,000 repetitions of the known process. The vertical reference is the true slope, 0.5.

The empirical SD of the slopes estimates their repeated-sample spread. Each fitted SE tries to estimate that spread using just one sample. Coverage is a property of the repeated interval-producing procedure, not a probability assigned to a fixed parameter after one interval is realized. The empirical coverage will not be exactly .95: at true .95 coverage, its Monte Carlo SE with 2,000 independent repetitions is $\sqrt{.95(.05)/2000} \approx 0.0049$.

The theoretical slope SD is $\sigma/\sqrt{S_{xx}}$. More noise increases it; greater predictor spread decreases it. More observations typically decrease it when the predictor distribution is held comparable. Table 8 changes one feature at a time and applies the formula to each design.

Table 8

How sample size, predictor support, and error SD affect theoretical slope precision.

Condition	n	Lower	Upper	Error SD	Slope SD
Baseline	40	6.0	16.0	1	0.0534
More rows	160	6.0	16.0	1	0.0272
Narrower support	40	8.5	13.5	1	0.1068
More noise	40	6.0	16.0	2	0.1068

Halving the predictor range at the same sample size doubles the slope SD; doubling σ also doubles it. Quadrupling the number of equally spaced points over the same range approximately halves it; the small difference from an exact half comes from the finite grids' variances. Using `x <- 1:n` while increasing n would change both sample size and predictor range. More rows do not automatically remove confounding, measurement error, dependence, or misspecification. These simulations establish behavior only under the mechanisms programmed into them.

5 Distinguish a Mean Response from a New-Case Prediction

Slope uncertainty concerns one process coefficient. Prediction introduces a different distinction: estimating a conditional mean versus anticipating one new outcome from the same declared process.

5.1 A fitted mean is not a new occupation

At $x_0 = 12$, the fitted income is 7.9322 thousand dollars. Under the model, this estimates the conditional mean for occupations with that average education. The standard error for the fitted mean is

$$SE(\hat{y}_0) = s_e \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

Substitution gives

$$3.4834 \sqrt{\frac{1}{102} + \frac{(12 - 10.738)^2}{751.8852}} = 0.3803.$$

The resulting 95% mean interval is $7.9322 \pm 1.984(0.3803) \approx [7.1776, 8.6868]$.

A new occupation's outcome also varies around its process mean. Write $\mu_0 = \beta_0 + \beta_1 x_0$, $\hat{\mu}_0 = \hat{y}_0$, and $Y_0 = \mu_0 + \varepsilon_0$. The prediction error has two parts:

$$Y_0 - \hat{\mu}_0 = \varepsilon_0 - (\hat{\mu}_0 - \mu_0).$$

If this new error is independent of the fitted data and comes from the same common-variance process, the two variance contributions add:

$$\text{Var}(Y_0 - \hat{\mu}_0 \mid \mathbf{X}, x_0) = \sigma^2 + \text{Var}(\hat{\mu}_0 \mid \mathbf{X}, x_0).$$

The first term is new outcome variation; the second is uncertainty in the estimated mean. Substituting the fitted-mean variance and estimating σ by s_e gives

$$SE_{\text{new}} = s_e \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = 3.5041.$$

Thus the 95% prediction interval is approximately $7.9322 \pm 1.984(3.5041) \approx [0.9802, 14.8842]$. The extra 1 is the new outcome's own error variance, expressed as a multiple of σ^2 ; it does not disappear as the estimation sample grows.

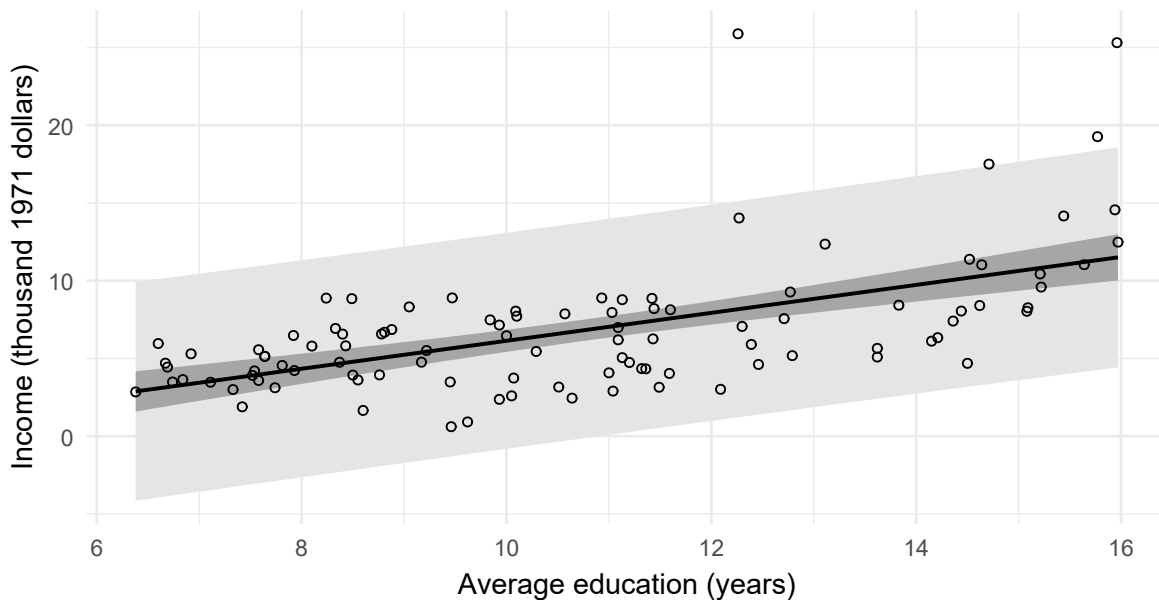


Figure 8: Mean-response intervals and new-case prediction intervals under the same working model.

Both intervals get wider away from $\bar{x}$, even within the observed range. Outside that range, numerical output is not a warrant for extrapolation. A lower prediction bound below zero also exposes the limitations of an unrestricted normal-error model for a nonnegative outcome; it is not a plausible negative-income forecast.

```
predict(model, data.frame(education = 12), interval = "confidence")
#>           fit           lwr           upr
#> 1 7.93216852 7.17757965 8.686757389
predict(model, data.frame(education = 12), interval = "prediction")
#>           fit           lwr           upr
#> 1 7.93216852 0.9801724429 14.8841646
```

A separately owed new-case prediction answer is a separable question, not automatically another strand of the original descriptive question. Moreover, a model prediction interval is not a test of a learning algorithm on unseen records. Held-out prediction performance is a distinct estimand, developed with multiple regression. Appendix A retains the interval derivations and further supported-value comparisons.

6 Check Form, Spread, Influence, and Support

6.1 Checks should change an interpretation or a decision

The fitted line does not have to satisfy a stochastic model to exist. But claims that it describes a linear conditional mean, that its classical interval has exact coverage, or that it predicts new occupations reliably need additional support.

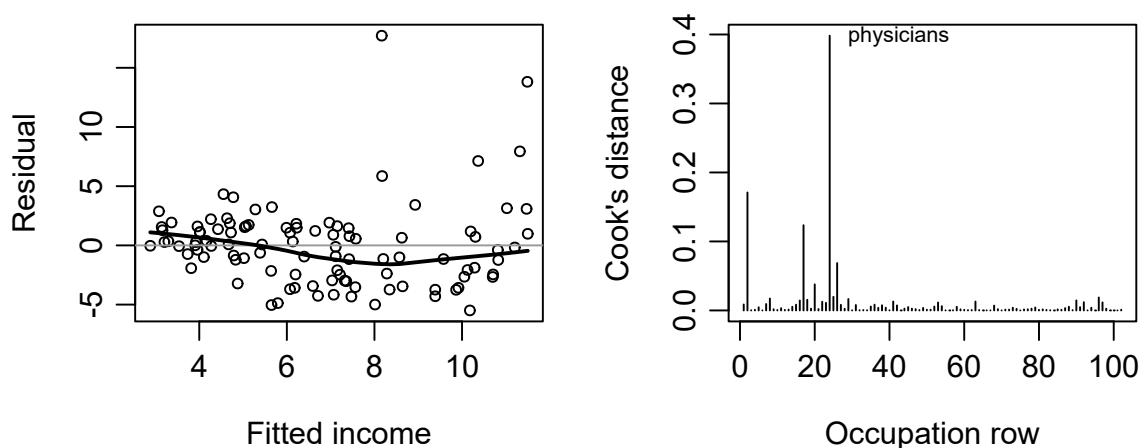


Figure 9: Residual pattern and influence invite different follow-up questions.

A curve in residuals suggests that one straight conditional mean leaves systematic variation. A changing residual spread challenges a common-variance uncertainty calculation. Leverage concerns unusual predictor positions; influence concerns how much the fitted model changes when a record's contribution changes. A large residual, high leverage, and high influence are related but different properties.

For a straight line with an intercept, leverage is

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

It grows as a predictor value moves away from $\bar{x}$. Cook's distance then combines leverage and residual magnitude:

$$D_i = \frac{e_i^2}{2s_e^2} \frac{h_i}{(1 - h_i)^2}.$$

Cook's distance summarizes how much the fitted values change when case i is omitted, relative to residual variation and the two fitted coefficients. It is a prompt to inspect provenance and sensitivity, not an automatic deletion cutoff.

For these data, adding education squared lets the fitted relationship bend. The following same-record comparison makes the change in shape visible. It is an exploratory check, not a retrospectively prespecified new final model.

```
quad <- lm(income_k ~ education + I(education^2), data = prestige)
most <- which.max(cooks.distance(model))
without <- lm(income_k ~ education, data = prestige[-most, ])
c(full = unname(coef(model)[2]), without = unname(coef(without)[2]))
#>          full          without
#> 0.8988128418 0.7982204353
coef(summary(quad))["I(education^2)", ]
#>      Estimate      Std. Error      t value      Pr(>|t|)
#> 0.134179313419 0.047636405684 2.816738825952 0.005856544055
```

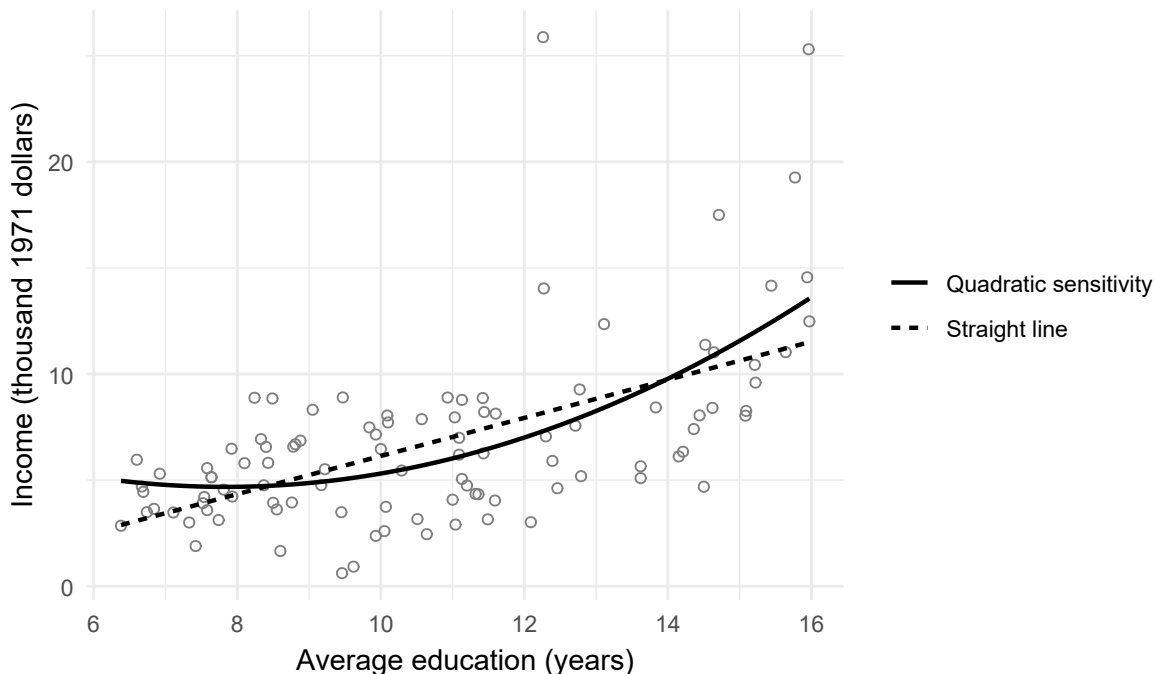


Figure 10: Straight and quadratic fits to the same 102 records. The curve is an exploratory sensitivity, not a selected final model.

The quadratic coefficient has $p = .0059$ under that model's classical reference calculation. This is a reason to examine the visible curvature, not a sufficient model-selection rule: unequal spread and choosing the comparison after inspecting the records also qualify that number.

Removing physicians, the record with the largest Cook's distance (0.3983), changes the slope from 0.8988 to 0.7982. Nothing in that result proves physicians were recorded incorrectly. The deletion is a sensitivity calculation; it is not our replacement analytic sample or a licence to hide a valid unusual occupation.

An HC3 heteroskedasticity-consistent covariance calculation gives a slope SE of 0.1491 rather than 0.1270. Using the same t_{100} critical value as an approximate sensitivity procedure gives [0.6031, 1.1945]. The line does not change. This procedure relaxes a variance assumption under suitable independent-observation conditions; it does not repair curvature, dependence, confounding, or measurement error (Long & Ervin, 2000; MacKinnon & White, 1985; White, 1980; Zeileis, 2004). Its finite-sample interval is not the exact classical normal-model interval. The short R calculation makes the covariance choice explicit:

```
V_hc3 <- sandwich::vcovHC(model, type = "HC3")
c(classical = coef(summary(model))["education", "Std. Error"],
  HC3 = sqrt(V_hc3["education", "education"]))
#>      classical      HC3
#> 0.1270354112 0.1490510028
```

Later chapters take these questions further. Chapter 8, Appendix B, assembles the HC3 matrix from its parts. Chapter 9, Appendix A, defines the target that a least-squares line estimates when the conditional mean is not linear, and its Section 5.3 shows that measurement error in the predictor attenuates a slope, which no covariance choice repairs. Nothing in this chapter relies on any of them.

Table 9

Diagnostic questions and their consequences.

Check	What needs to be examined	Consequence
Form	Residual curve and a curved-fit comparison	A straight line may remain a description but fail as a constant conditional-mean slope account
Spread	Residual variation over fitted values	Classical SEs may need revision; HC3 is a variance sensitivity, not a universal repair
Influence	Source record and result with/without a consequential case	Disclose sensitivity; correct errors only when verified
Independence	How occupations and measurements were generated or related	A plot cannot establish independent process realizations
Support	Actual education range and density of observations	Restrict comparisons; avoid extrapolation and individual-level claims

7 Regression Toward the Mean

7.1 A line in standard units

Francis Galton, who gave regression its name, measured the adult heights of parents and their grown children in 205 families in the 1880s (Galton, 1886). We use James Hanley’s transcription of Galton’s notebook (Hanley, 2004), distributed in the R package `HistData` (Friendly et al., 2025), and keep one row per family: among the 179 families with at least one son, one son was drawn at random from each, so every father appears once and the rows describe separate families. Heights are in inches, mostly recorded to the nearest half inch. The small extract `galton1886_sons.csv` is supplied with the chapter under this note: *Galton’s family heights (published 1886), transcribed by J. A. Hanley and distributed in the R package HistData under the GPL; one son per family drawn for QMSBR. The measurements are historical records in the public domain; the extract is redistributed with this attribution.*

Chapter 2 measured a value’s distance from its mean in standard units, $z_x = (x - \bar{x})/s_x$, and Section 2.2 wrote the slope as $\hat{\beta}_1 = r s_y/s_x$. Substituting that slope into the line through the mean point, $\hat{y} - \bar{y} = \hat{\beta}_1(x - \bar{x})$, and dividing both sides by s_y gives the line in standard units:

$$\frac{\hat{y} - \bar{y}}{s_y} = r \frac{x - \bar{x}}{s_x}, \quad \text{that is,} \quad \hat{z}_y = r z_x.$$

The line passes through the origin with slope r . Unless the correlation is perfect, $|r| < 1$, so a case z_x SDs from the mean on the predictor is predicted only $r z_x$ SDs from the mean on the outcome, each measured in its own SDs. Galton called this regression toward mediocrity; we call it **regression toward the mean**.

```
galton <- read.csv("galton1886_sons.csv")
sapply(galton[c("father", "son")], sd)
#>      father      son
#> 2.546555039 2.647465487
c(r = cor(galton$father, galton$son),
  z_slope = unname(coef(lm(scale(son) ~ scale(father),
    data = galton))[2]))
#>      r      z_slope
#> 0.4269305719 0.4269305719
```

`sapply()` applies `sd()` to both columns, and `scale()` converts a column to standard units. The fathers average 69.0989 inches with SD 2.5466, the sons 69.2140 inches with SD 2.6475, and their correlation is $r = .4269$. In inches the fitted slope is 0.4438 inch of son’s height per inch of father’s height; fitted to the standardized heights it is .4269, which is r , as the equation

says. A father 1.6 SD above the fathers' mean has a son predicted $.4269 \times 1.6 = 0.6831$ SD above the sons' mean: still taller than average, but less than half as far out.

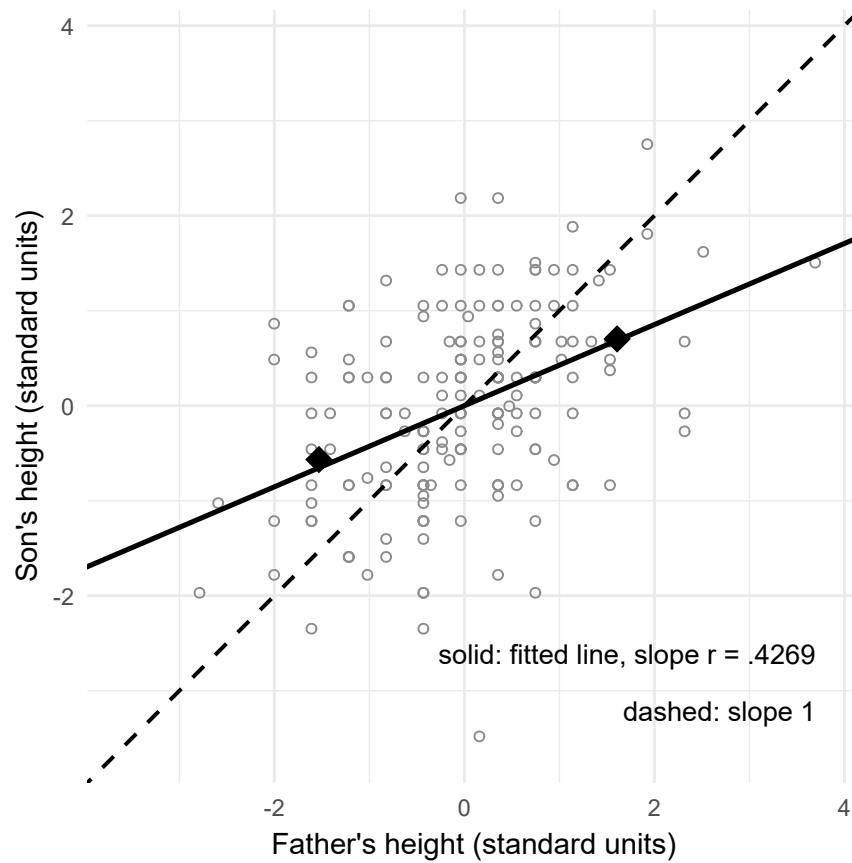


Figure 11: Galton's 179 fathers and sons in standard units. The fitted line has slope $r = .4269$, flatter than the dashed line of slope 1; the filled points are the mean positions of the fathers more than 1 SD above and below their mean, with their sons.

In Figure 11 the dashed line of slope 1 marks where a son would sit if he were exactly as many SDs from his mean as his father. The fitted line is flatter. To the right of the mean it lies below the dashed line and to the left above it: tall fathers have sons predicted less tall, and short fathers sons predicted less short, in standard units. The two filled points mark the average positions of the fathers more than 1 SD above and below their mean, with their sons; both sit near the fitted line, well inside the dashed one.

7.2 Extreme groups move toward the mean; the spread does not

Table 10 groups the men in bands of 1 SD, first by the fathers' heights and then by the sons'. Each band includes its upper bound, so the four bands of each block hold all 179 men.

Table 10

Groups of Galton's fathers and sons in standard units, selected on either height.

Selected on	Band (SDs)	Men	Mean z of the selected height	Mean z of the other height	r times the selected mean
Fathers	Above +1	24	1.6072	0.7029	0.6862
Fathers	0 to +1	60	0.4978	0.2251	0.2125
Fathers	-1 to 0	65	-0.3463	-0.2058	-0.1479
Fathers	Below -1	30	-1.5310	-0.5668	-0.6536
Sons	Above +1	31	1.4057	0.6224	0.6001
Sons	0 to +1	60	0.4650	0.1431	0.1985
Sons	-1 to 0	61	-0.4610	-0.0697	-0.1968
Sons	Below -1	27	-1.6057	-0.8751	-0.6855

Read the first block row by row. The 24 fathers more than 1 SD above their mean average $z = 1.6072$, or 73.1917 inches; their sons average 0.7029, or 71.0750 inches, close to the line's prediction $.4269 \times 1.6072 = 0.6862$. The 30 short fathers mirror them. In every row the other height's mean lies between zero and the selected mean, near r times it; bands nearer the middle move less, because they start nearer the mean.

The second block selects on the sons, and the movement toward the mean appears again, now running backward in time: the 31 sons more than 1 SD above their mean, at 1.4057, had fathers averaging only 0.6224. No force pulls heights toward the average from one generation to the next; such a force could not act backward in time. An imperfect correlation is enough for the fitted line: its predictions lie closer to the mean, in standard units, than the values they are predicted from. For groups of men, the pattern also needs the mean of one height to follow a straight line in the other, as it roughly does in these families; then selecting men for an extreme value of one height selects, on average, men whose other height is less extreme. On average is the operative phrase: a group's observed mean scatters around its prediction, so an imperfect correlation does not guarantee that every group selected for an extreme value sits nearer the mean on the other height.

If tall fathers have sons nearer the mean and short fathers do too, should the sons' heights be less spread out than their fathers'? They are not: the sons' SD, 2.6475 inches, is slightly larger than the fathers', 2.5466. This is the **regression paradox**. The fitted values carry only the part of a son's height that his father's height predicts, and they spread only $r s_y = 1.1303$ inches. The residuals carry the rest, with spread $s_y \sqrt{1 - r^2} = 2.3941$ inches (both SDs with the $n - 1$ divisor; `sigma()` divides by $n - 2$ and gives 2.4008). The two parts combine through

their variances, not their SDs: $1.1303^2 + 2.3941^2 = 2.6475^2$, Section 2.6's decomposition of total variation, in which the fitted part holds the share $R^2 = r^2 = .1823$. Some sons of average fathers are very tall or very short, so the sons' generation is as varied as the fathers'. Regression toward the mean describes predictions for selected groups, not a population growing more uniform.

7.3 The regression fallacy

Regression toward the mean becomes a fallacy when the movement it predicts is credited to a cause. Tversky and Kahneman (1974) recount a conversation with flight instructors: a trainee praised for an unusually smooth landing tended to land less well next time, and a trainee reprimanded for a rough landing tended to do better, so the instructors concluded that praise hurts and criticism helps.

The line in standard units explains the pattern without either conclusion. A landing's quality reflects the trainee's skill and the chance conditions of that attempt, so two landings by the same trainee are correlated, but not perfectly. The landings that drew comment were selected for being extreme, and the next landing is predicted nearer the trainee's usual level whatever the instructor says: the praise and the criticism coincided with regression toward the mean. Crediting that movement to the feedback is the **regression fallacy**.

The same pattern appears whenever cases are selected for an extreme first measurement and measured again: pupils chosen for their low test scores and given extra help, patients enrolled when their symptoms peak, schools placed on a list after their worst year. On the second measurement such groups move toward the mean on average even if nothing done in between has any effect. A comparison group selected the same way and measured at the same two times shows how much movement selection alone produces. Assigning the help at random among the selected cases does better still, since both groups then regress alike and their difference estimates the help's effect (Barnett et al., 2005). Chapter 14, Section 4.1, applies this reasoning to selection on a noisy baseline.

8 Reading Bridge

A simple regression starts with a comparison and a unit. Its fitted values and residuals are calculations on records. Least squares connects the slope to covariance and predictor variation and puts the line through the mean point. Residual SD, R^2 , slope uncertainty, and new-case variation answer different questions; the matched Pearson, slope, and F statistics connect calculations without supplying a sampling design.

Across the 102 recorded occupations, fitted income rises by 0.8988 thousand 1971 Canadian dollars—about 899 dollars—per additional year of occupation-average education over the observed 6.38–15.97-year range. This is a bounded description of the records, not an individual,

causal, or population return. Known-truth repetition showed how a separately declared process gives slope uncertainty and coverage. In standard units the line predicts r times a case's distance from the mean, so groups selected for extreme values are predicted nearer the mean on another measurement while the spread stays the same; a before-and-after gain in such a group needs a comparison selected the same way before it is read as an effect. Supplementary A reads a short report of the occupation line as an argument, locating its reasons before judging its answer.

Chapter 8 refits the education-only line on the same 98 type-complete occupations before adding prestige, percentage of women, and occupation type. Moving from 102 to 98 changes the records; it is not an adjustment effect. The chapter develops conditional comparisons and standardized fitted quantities, and asks a separate held-out prediction question. Chapter 9 collects the regression toolbox these chapters use only briefly, from factors with several levels to robust and resampled uncertainty. Chapter 9, Section 7 (Project STAR), then reads a randomized class-size experiment whose uncertainty follows its school-clustered design (Finn & Achilles, 1990; Mosteller, 1995); Chapter 10 extends the factorial interactions from Chapters 4 and 6 to comparisons involving regression predictors.

Key terms

Use this short reference to revisit the regression concepts introduced here. The Part One Concept Reference gives the broader reference and the principal home of each concept; fitted values, residuals, standard errors, intervals, standardized values, and the reading vocabulary are explained in Chapters 1–4.

Table 11

Selected concepts for simple linear regression.

Term and common alternatives	Meaning
Simple linear regression (regression line)	A straight-line summary of how one outcome differs across one predictor: fitted value equals intercept plus slope times the predictor. Its descriptive use concerns the records fitted; a model-based use needs a declared process.
Intercept	The fitted outcome at predictor value zero. It places the line vertically; when zero is outside the observed support, it is a mathematical anchor rather than a supported prediction.
Slope (regression coefficient)	Centering the predictor moves it to a supported reference point. The difference in fitted outcome per one-unit difference in the predictor, in outcome units per predictor unit. It compares units that differ in the predictor; it is not a change within one unit unless a design supports that reading.

Term and common alternatives	Meaning
Ordinary least squares (OLS)	The rule that chooses the intercept and slope minimizing the residual sum of squares. With an intercept, the fitted line passes through the mean point and the residuals sum to zero; the slope equals covariance divided by predictor variance.
Residual sum of squares (RSS)	The sum of squared residuals from a fitted line, in squared outcome units. Total squared variation about the mean splits into a regression part and RSS when the model has an intercept.
Residual standard deviation (residual standard error)	The square root of RSS divided by its residual degrees of freedom, in outcome units; with an intercept and one predictor the degrees of freedom are $n - 2$. Under the process model it estimates the error SD.
R-squared (coefficient of determination)	One minus RSS divided by total squared variation: the proportion of centered squared outcome variation accounted for by the fit in these records. In simple regression with an intercept it equals the squared correlation.
Confidence interval for a mean response (mean-response interval)	An interval for the process mean outcome at a stated predictor value, using the SE of the fitted value. It concerns an average, narrows near the predictor mean, and does not cover individual outcomes.
Prediction interval (new-case interval)	An interval for one new outcome at a stated predictor value from the same process. Its variance adds the error variance to the fitted-mean variance, so it is wider than the mean-response interval and keeps a nonzero floor as the sample grows.
Leverage	A measure of how unusual a record's predictor value is relative to the others; it grows with distance from the predictor mean and appears in the SE of a fitted value and in influence measures.
Cook's distance (influence)	A summary of how much the fitted values change when one record is omitted, combining its residual and its leverage. It prompts inspection of provenance and a disclosed sensitivity, not automatic deletion.
Heteroskedasticity-consistent covariance (HC3; sandwich estimator)	A coefficient covariance estimate that allows unequal error variance across predictor values under independence. It changes SEs and intervals, not the fitted coefficients, and does not repair curvature, dependence, measurement error, or confounding.
Extrapolation (predictor support)	Using a fitted line at predictor values outside the observed range or where observations are sparse. Software returns a number there; the data do not support it.

Term and common alternatives	Meaning
Regression toward the mean (regression to the mean)	In standard units a case's predicted outcome is r times its predictor value, $\hat{z}_y = r z_x$, so with $ r < 1$ cases extreme on one measurement are predicted nearer the mean on the other, and a group selected for an extreme noisy first value is expected to move toward the mean on a second measurement, on average rather than in every group. The outcome's spread does not shrink (the regression paradox).
Regression fallacy	Attributing to a treatment, a talent or another cause the movement toward the mean that selection on an extreme noisy measurement produces by itself. A comparison group selected the same way, or random assignment, separates a real change from regression toward the mean.

Further reading

For another accessible account of lines, residuals, and fit, see OpenIntro's single-predictor regression tutorial. For a compact executable route from a fitted model to intervals and checks, see the ISLR authors' Chapter 3 R lab (James et al., 2021). These provide complementary practice; the present chapter additionally makes the finite-record and model-based questions explicit. Consult the carData reference manual (Fox et al., 2022) for data documentation, Barnett et al. (2005) on regression to the mean in repeated measurements, and Fox (2016), Fox & Weisberg (2019), and Gelman et al. (2020) for broader regression treatments.

Exercises

The prompts below use **Foundational** and **Integrative** levels. Where rounded summaries are supplied, answers within 0.02 in the stated units are sufficient. Exercises 1–6 draw on the main chapter.

Exercise 1: Read the unit and the line (Foundational)

A teaching report describes 80 occupations with $\widehat{\text{income}} = -1.5 + 0.7 \text{ education}$, where income is in thousand dollars and average education ranges from 7 to 16 years. Interpret the slope and the intercept. Calculate the fitted income at 12 years and the residual for an occupation at 12 years with income 9. Explain why the slope does not give an individual's gain from another year of schooling.

Exercise 2: Connect covariance and the fitted line (Foundational)

A set of records has $\bar{x} = 4$, $\bar{y} = 10$, $S_{xx} = 20$, $S_{xy} = 30$. Calculate the OLS slope and intercept, and the fitted value at $x = 5$. Show that the line passes through the mean point. If $s_x = 2$ and $s_y = 5$, calculate the correlation implied by these summaries. What would prevent fitting a slope if every x were equal?

Exercise 3: Separate fit and uncertainty (Foundational)

An intercept-and-slope regression has $n = 22$, $RSS = 80$, $TSS = 144$, $S_{xx} = 100$, and fitted slope 0.8. Calculate residual SD, R^2 , and the classical slope SE. Use $t_{.975,20} = 2.086$ to obtain a 95% slope interval. State the process assumptions needed for that interval and explain why it does not by itself establish that the line fits closely or identifies a cause.

Exercise 4: Mean or new case? (Foundational)

Under the stated independent normal-error constant-variance model, a fitted value at $x_0 = \bar{x}$ is 12, $n = 25$, and $s_e = 3$. Use $t_{.975,23} = 2.069$. Calculate the SE and 95% interval for the conditional mean, then the corresponding prediction SE and interval for one independent new case at the same x_0 . Explain the extra 1 in the prediction expression. Would either interval validate the model in a different population?

Exercise 5: Tell regression toward the mean from an effect (Integrative)

In an invented study, a school records a reading test in September and again in June for all 200 of its pupils. Both tests have mean 50 and SD 10, and their correlation is .6. The 20 pupils with the lowest September scores, averaging 32 points, receive extra tutoring, and their June mean is 42.8 points. These figures include the tutored pupils' June scores, so by themselves they cannot show what those scores would have been without tutoring. The school's records from earlier years, when no pupil was tutored, supply that benchmark: the two tests again had means 50, SDs 10 and correlation .6, and the mean June score of pupils with the same September score followed a straight line in that score. Assume that this year's pupils would have followed the same pattern without tutoring. (1) Express the tutored group's September mean in standard units, and give the June mean that this pattern's line in standard units predicts for pupils with that September score, in standard units and in points. (2) The school reports that tutoring raised the group by 10.8 points. Name the error in reading the whole gain as an effect of tutoring, and say how much of the gain regression toward the mean predicts by itself under this pattern. (3) The report adds that, since the lowest scorers caught up, the June scores of all 200 pupils must be less spread out than September's. Split a June score into the part that the September score predicts and the part it does not, give the SD of each part, and explain why regression toward the mean leaves the June SD at 10. (4) Describe a comparison that would separate an effect of tutoring from regression toward the mean.

Exercise 6: Respond to sensitivity (Integrative)

In a descriptive analysis of 102 occupations, the slope is 0.9. A valid, high-income occupation has the largest Cook's distance; removing it gives 0.79. A curved-fit check changes the shape, and an HC3 calculation changes the slope SE from 0.13 to 0.15 without changing the line.

What does each check tell you? What does it not tell you? Explain why deleting the occupation automatically is inappropriate. Distinguish a variance repair for the same estimand from replacing a constant process slope with a different estimand. Write a short sensitivity disclosure for the descriptive analysis.

Appendix A: Extended regression mathematics and interval derivations

The main text uses the identities needed to read a fitted line. This appendix derives them and distinguishes identities about the observed records from claims about hypothetical repetitions. All occupation calculations retain the same 102 rows, education in years, and income in thousand 1971 Canadian dollars.

Derivatives, normal equations, and the minimizing line

Start with a fitted model that predicts one common center m for every observation. Chapter 2 Section 2.2 states that the mean minimizes the sum of squared deviations, and Chapter 3 Appendix A derives the identity behind it:

$$L(m) = \sum_i (y_i - m)^2 = \sum_i (y_i - \bar{y})^2 + n(m - \bar{y})^2.$$

Write $y_i - m = (y_i - \bar{y}) + (\bar{y} - m)$ and expand the square. The cross term is zero because $\sum_i (y_i - \bar{y}) = 0$. For $n > 0$, the last term is nonnegative and vanishes only at $m = \bar{y}$, so the mean uniquely minimizes this criterion.

The three outcomes 2, 4, 3 from Section 2.3 have mean 3 and $L(m) = 2 + 3(m - 3)^2$, so their best constant line has loss 2. Allowing the fitted value to depend on the three predictor values reduces RSS to 1.5, as Table 3 showed. Neither result requires Normal errors or a probability sample. Chapter 6's likelihood example chose a parameter by optimizing a different stated criterion; here we develop the least-squares line algebraically. An uncertainty or causal claim still needs its own supporting conditions.

The intercept-and-slope criterion has two coefficients to determine. Instead of setting one derivative to zero, we set its two partial derivatives to zero and verify that the resulting line is the minimum.

For candidate intercept b_0 and slope b_1 , define

$$Q(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Squared residuals cannot cancel and give large vertical discrepancies greater weight. Differentiating with respect to each candidate coefficient gives

$$\frac{\partial Q}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \quad \frac{\partial Q}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

At the minimum both derivatives are zero. The resulting **normal equations** are

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i = \sum_i y_i, \quad \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2 = \sum_i x_i y_i.$$

The first equation gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Substitution into the second yields

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2} = \frac{S_{xy}}{S_{xx}}.$$

Here $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$ is required: a slope cannot be identified when every predictor value is the same. Because sample covariance and variance use the same denominator $n - 1$,

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = r \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

In the occupation data,

$$\hat{\beta}_1 = \frac{675.8041}{751.8852} \approx 0.8988, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx -2.8536.$$

The ratio has units of thousand dollars per education year. Unlike a correlation, the slope changes when a variable's units change. Standardizing both variables makes their sample standard deviations one, so the simple intercept-model slope becomes r ; Section 7 reads that line.

To verify that the stationary point is the unique minimum, write any other line as $\hat{\beta}_0 + d_0 + (\hat{\beta}_1 + d_1)x_i$. The normal equations make the cross term with the OLS residuals vanish, leaving

$$Q(\hat{\beta}_0 + d_0, \hat{\beta}_1 + d_1) = RSS + n(d_0 + d_1 \bar{x})^2 + d_1^2 S_{xx}.$$

With $S_{xx} > 0$, every different line has larger RSS. If a candidate line is constrained to pass through $(\bar{x}, \bar{y})$, its excess RSS is simply $S_{xx}(b_1 - \hat{\beta}_1)^2$. Section 2.4 makes this comparison visible with five candidate lines.

Residual constraints and projection geometry

At the OLS solution the derivative equations become $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$. Thus the residuals have zero sample covariance with the predictor and mean zero **by construction**. They are not evidence that the unknown population errors are independent, normally distributed, or conditionally mean zero.

For the geometry, let $\mathbf{X}$ be the $n \times 2$ matrix with first column all ones and second column containing the predictor. Bold $\mathbf{X}$ is a matrix, distinct from a single predictor value x_i . The normal equations can be written

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}, \quad \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

The fitted vector is the projection of $\mathbf{y}$ onto the two-dimensional space spanned by the intercept and predictor columns. The residual vector is perpendicular to that space. Writing $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, we have $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

Because the model includes an intercept, both $\hat{\mathbf{y}}$ and the constant vector $\bar{y}\mathbf{1}$ lie in the fitted space. Consequently,

$$\begin{aligned} \sum_i (y_i - \bar{y})^2 &= \sum_i \{(\hat{y}_i - \bar{y}) + e_i\}^2 \\ &= \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i e_i^2, \end{aligned}$$

since $\sum_i (\hat{y}_i - \bar{y})e_i = 0$. This proves $TSS = RegSS + RSS$, an algebraic decomposition of observed variation, not a division into causal contributions. For these records, $1820.8134 = 607.4214 + 1213.3920$, in squared thousand-dollar units.

Also $RegSS = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$. Dividing by $TSS = S_{yy}$ proves $R^2 = S_{xy}^2 / (S_{xx}S_{yy}) = r^2$ for simple regression **with an intercept**. The identities need not hold for a no-intercept fit. When fitted values are nonconstant, $\text{cor}(y, \hat{y})^2 = R^2$ as well.

There are two fitted directions and $n - 2$ residual directions. Under a correct linear mean with independent constant-variance errors, $E(RSS | \mathbf{X}) = (n - 2)\sigma^2$. Thus $s_e^2 = RSS / (n - 2)$ estimates the error variance, while s_e estimates its standard deviation in the outcome's units. The square root is not itself an exactly unbiased estimator of σ . Without that stochastic model, s_e remains a computable residual-scale summary, not proof of a common population error variance.

From an error model to slope and intercept uncertainty

The 102-record least-squares line is already determined by those records. The next derivation concerns a **separate model-based question**: what constant conditional-mean coefficient generated hypothetical independent occupation outcomes at the observed education values? Assume

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad E(\varepsilon_i \mid \mathbf{X}) = 0, \quad \text{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}.$$

These are offered working conditions, not consequences of the package's availability or of fitting `lm()`. The slope error is

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x}) \varepsilon_i}{S_{xx}}.$$

Independence and common variance then give

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sigma^2}{S_{xx}}, \quad SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{S_{xx}}}.$$

Because $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ and $\text{Cov}(\bar{Y}, \hat{\beta}_1 \mid \mathbf{X}) = 0$,

$$\text{Var}(\hat{\beta}_0 \mid \mathbf{X}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{X}) = -\frac{\bar{x} \sigma^2}{S_{xx}}.$$

This covariance explains why simply adding the variances of an uncentered intercept and slope prediction would be wrong. If the predictor is centered at its observed mean, the intercept and slope estimates are uncorrelated under the classical model, and the intercept SE is $s_e/\sqrt{n}$.

Under the stated linear-mean, uncorrelated, equal-variance conditions, OLS is the best linear unbiased estimator: it has the smallest variance among estimators linear in the outcomes and unbiased for the coefficients. This Gauss–Markov statement does not require normal errors and does not establish causal identification, valid measurement, or population coverage (Fox, 2016; Kutner et al., 2005).

Add a jointly normal error vector conditional on the predictor values. Then the standardized coefficient error is normal, RSS/σ^2 has a χ^2_{n-2} distribution, and these quantities are independent. Their ratio gives

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-2}, \quad j = 0, 1.$$

That is the basis of exact classical conditional coverage for $\hat{\beta}_j \pm t_{.975, n-2} SE(\hat{\beta}_j)$, and of the corresponding null test. For the occupation slope the calculation is $0.8988 \pm 1.984(0.127) = [0.6468, 1.1508]$ thousand dollars per year. It illustrates the model-based procedure; the diagnostics in Section 6 do not support treating its nominal exact coverage as established here.

Fixed- X derivations condition on the predictor values; those values need not literally have been experimentally fixed. A random- X conditional-mean model can use the same conditional argument when its error conditions hold. Neither route demonstrates that these occupations were probability sampled. Normality of the marginal distribution of education or income is not required. The assumptions concern conditional errors, not the two marginal histograms.

Why a new-case interval has an additional variance term

For a supported value x_0 , define the model mean $\mu_0 = \beta_0 + \beta_1 x_0$ and its estimate $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. Substituting the coefficient variances and covariance above gives

$$\begin{aligned} \text{Var}(\hat{\mu}_0 \mid \mathbf{X}) &= \text{Var}(\hat{\beta}_0) + x_0^2 \text{Var}(\hat{\beta}_1) + 2x_0 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2 \left\{ \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\}. \end{aligned}$$

The common conditioning on $\mathbf{X}$ is suppressed in the first line. A confidence interval for this mean uses the square root of this expression with s_e replacing σ .

A new occupation outcome is instead $Y_0 = \mu_0 + \varepsilon_0$. If its error is independent of the fitted data and comes from the same normal, equal-variance process,

$$\text{Var}(Y_0 - \hat{\mu}_0 \mid \mathbf{X}, x_0) = \sigma^2 \left\{ 1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right\}.$$

The extra 1 is new outcome variation; it does not disappear even if the model mean were known. Both standard errors use the same $t_{.975, n-2}$ critical value for their classical 95% intervals under the full model.

The table applies both formulas at four supported education values, the same calculation as Section 5's two `predict()` calls, and puts both sets of limits beside the common fitted values.

Table 12*Mean-response and new-case intervals at four supported education values.*

Education	Fitted	Mean lower	Mean upper	Case lower	Case upper
8	4.3369	3.3651	5.3087	−2.6420	11.3158
10	6.1345	5.4254	6.8437	−0.8127	13.0818
12	7.9322	7.1776	8.6868	0.9802	14.8842
15	10.6286	9.3550	11.9022	3.6013	17.6559

At 10 years, the point prediction is 6.1345 thousand dollars, the mean interval is [5.4254, 6.8437], and the new-occupation interval is [−0.8127, 13.0818]. The negative endpoint exposes the lower-tail limitation of this unbounded normal-linear model. Cutting it off at zero does not produce a validated 95% interval. An outcome transformation or a model respecting the boundary would change the analysis and would need its own interpretation and checks.

Both intervals widen away from $\bar{x}$; their narrowest point is the mean education value. With more comparable independent observations and stable predictor spread, mean uncertainty tends to shrink. The new-case interval retains a nonzero error-variation floor. None of these formulas includes uncertainty from an inappropriate mean function, biased measurement, or a shift to another labour market. They are pointwise model intervals, not a simultaneous band or empirical out-of-sample validation. Their target is a new occupation-level record from the declared process, never an individual worker’s income.

References

- Barnett, A. G., van der Pols, J. C., & Dobson, A. J. (2005). Regression to the mean: What it is and how to deal with it. *International Journal of Epidemiology*, 34(1), 215–220. <https://doi.org/10.1093/ije/dyh299>
- Finn, J. D., & Achilles, C. M. (1990). Answers and questions about class size: A statewide experiment. *American Educational Research Journal*, 27(3), 557–577.
- Fox, J. (2016). *Applied regression analysis and generalized linear models* (3rd ed.). SAGE Publications.
- Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. <https://www.john-fox.ca/Companion/index.html>
- Fox, J., Weisberg, S., & Price, B. (2022). *carData: Companion to applied regression data sets* (Versions 3.0-5). Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.carData>
- Friendly, M., Dray, S., Li, P., & Bellhouse, D. (2025). *HistData: Data sets from the history of statistics and data visualization* (Version 1.0.0). Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.HistData>

- Galton, F. (1886). Regression towards mediocrity in hereditary stature. *The Journal of the Anthropological Institute of Great Britain and Ireland*, 15, 246–263. <https://doi.org/10.2307/2841583>
- Gelman, A., Hill, J., & Vehtari, A. (2020). *Regression and other stories*. Cambridge University Press. <https://doi.org/10.1017/9781139161879>
- Hanley, J. A. (2004). “Transmuting” women into men: Galton’s family data on human stature. *The American Statistician*, 58(3), 237–243. <https://doi.org/10.1198/000313004X1558>
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>
- Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.
- Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224. <https://doi.org/10.1080/00031305.2000.10474549>
- MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325. [https://doi.org/10.1016/0304-4076\(85\)90158-7](https://doi.org/10.1016/0304-4076(85)90158-7)
- Mosteller, F. (1995). The Tennessee study of class size in the early school grades. *The Future of Children*, 5(2), 113–127.
- Robinson, W. S. (1950). Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3), 351–357. <https://doi.org/10.2307/2087176>
- Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124–1131. <https://doi.org/10.1126/science.185.4157.1124>
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838. <https://doi.org/10.2307/1912934>
- Zeileis, A. (2004). Econometric computing with HC and HAC covariance matrix estimators. *Journal of Statistical Software*, 11(10), 1–17. <https://doi.org/10.18637/jss.v011.i10>